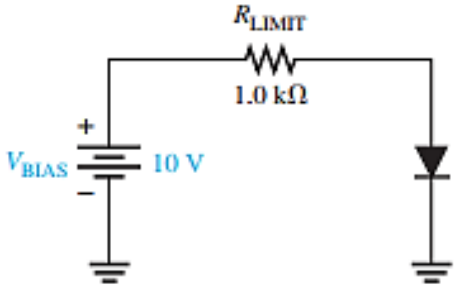
1 Determine the forward voltage and forward current for the diode in the figure below for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $r'_d = 10\ \Omega$ at the determined value of forward current.



Ideal model:

$$V_F = \mathbf{0\ V}$$

$$I_F = \frac{V_{BIAS}}{R_{LIMIT}} = \frac{10\ V}{1.0\ k\Omega} = \mathbf{10\ mA}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (10\ mA)(1.0\ k\Omega) = \mathbf{10\ V}$$

Practical model:

$$V_F = \mathbf{0.7\ V}$$

$$I_F = \frac{V_{BIAS} - V_F}{R_{LIMIT}} = \frac{10\ V - 0.7\ V}{1.0\ k\Omega} = \frac{9.3\ V}{1.0\ k\Omega} = \mathbf{9.3\ mA}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (9.3\ mA)(1.0\ k\Omega) = \mathbf{9.3\ V}$$

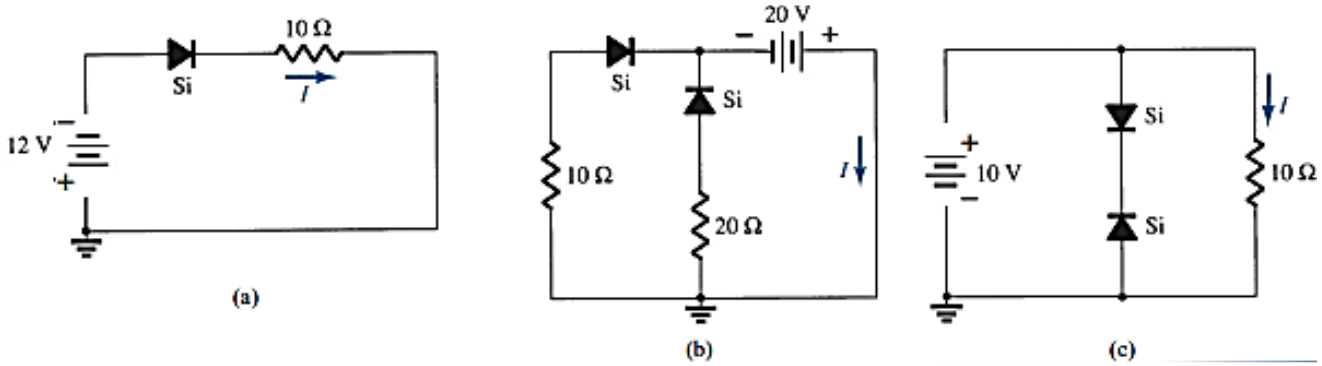
Complete model:

$$I_F = \frac{V_{BIAS} - 0.7\ V}{R_{LIMIT} + r'_d} = \frac{10\ V - 0.7\ V}{1.0\ k\Omega + 10\ \Omega} = \frac{9.3\ V}{1010\ \Omega} = \mathbf{9.21\ mA}$$

$$V_F = 0.7\ V \div I_F r'_d = 0.7\ V \div (9.21\ mA)(10\ \Omega) = \mathbf{792\ mV}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (9.21\ mA)(1.0\ k\Omega) = \mathbf{9.21\ V}$$

2 Determine the current I for each of the configurations using the approximate equivalent model for the diode.



(a) $I = \mathbf{0\ mA}$; diode reverse-biased.

(b) $V_{20\Omega} = 20\ \text{V} - 0.7\ \text{V} = 19.3\ \text{V}$ (Kirchhoff's voltage law)

$$I(20\ \Omega) = \frac{19.3\ \text{V}}{20\ \Omega} = 0.965\ \text{A}$$

$$V(10\ \Omega) = 20\ \text{V} - 0.7\ \text{V} = 19.3\ \text{V}$$

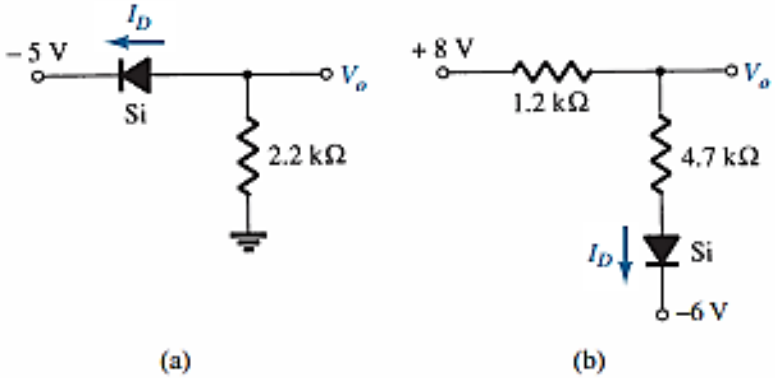
$$I(10\ \Omega) = \frac{19.3\ \text{V}}{10\ \Omega} = 1.93\ \text{A}$$

$$I = I(10\ \Omega) + I(20\ \Omega)$$

$$= \mathbf{2.895\ A}$$

(c) $I = \dfrac{10\ \text{V}}{10\ \Omega} = \mathbf{1\ A}$; center branch open

3 Determine V_o and I_D for the networks:



(a) (b)

(a) Diode forward-biased,

Kirchhoff's voltage law (CW): $-5\text{ V} + 0.7\text{ V} - V_o = 0$

$$V_o = \mathbf{-4.3\text{ V}}$$

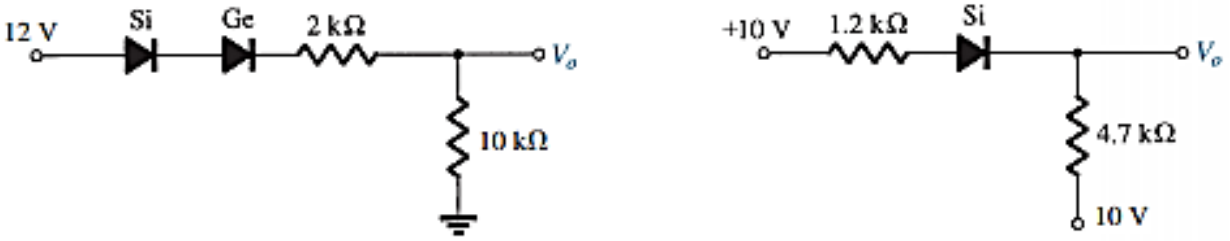
$$I_R = I_D = \frac{|V_o|}{R} = \frac{4.3\text{ V}}{2.2\text{ k}\Omega} = \mathbf{1.955\text{ mA}}$$

(b) Diode forward-biased,

$$I_D = \frac{8\text{ V} + 6\text{ V} - 0.7\text{ V}}{1.2\text{ k}\Omega + 4.7\text{ k}\Omega} = \mathbf{2.25\text{ mA}}$$

$$V_o = 8\text{ V} - (2.25\text{ mA})(1.2\text{ k}\Omega) = \mathbf{5.3\text{ V}}$$

4 Determine the level of V_o

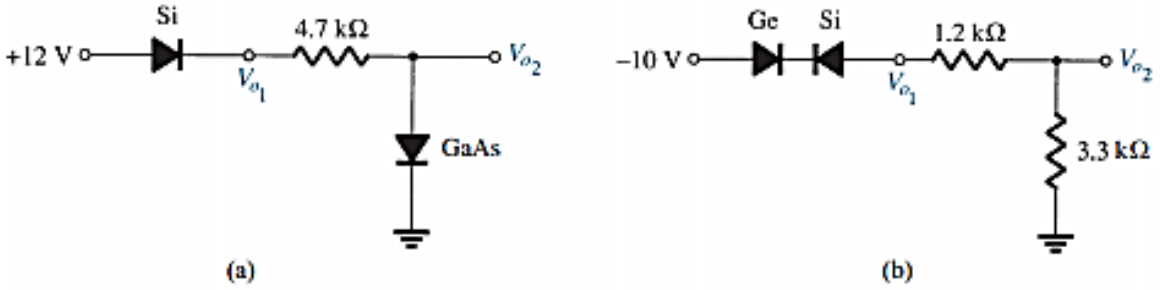


(a) (b)

(a) $$V_o = \frac{10\text{ k}\Omega(12\text{ V} - 0.7\text{ V} - 0.3\text{ V})}{2\text{ k}\Omega + 10\text{ k}\Omega} = \mathbf{9.17\text{ V}}$$

(b) $V_o = \mathbf{10\text{ V}}$

5 Determine V_{o1} and V_{o2} for the networks:



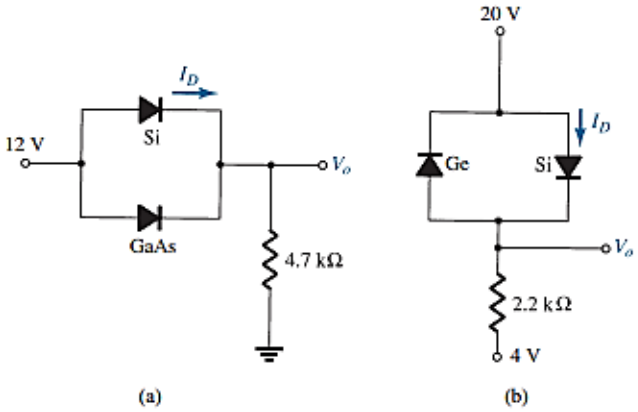
(a) $V_{o_1} = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$

$V_{o_2} = \mathbf{1.2\text{ V}}$

(b) $V_{o_1} = \mathbf{0\text{ V}}$

$V_{o_2} = \mathbf{0\text{ V}}$

6 Determine V_o and I_D for the networks:



(a) Both diodes forward-biased

Si diode turns on first and locks in 0.7 V drop.

$$I_R = \frac{12\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega} = 2.4\text{ mA}$$

$I_D = I_R = \mathbf{2.4\text{ mA}}$

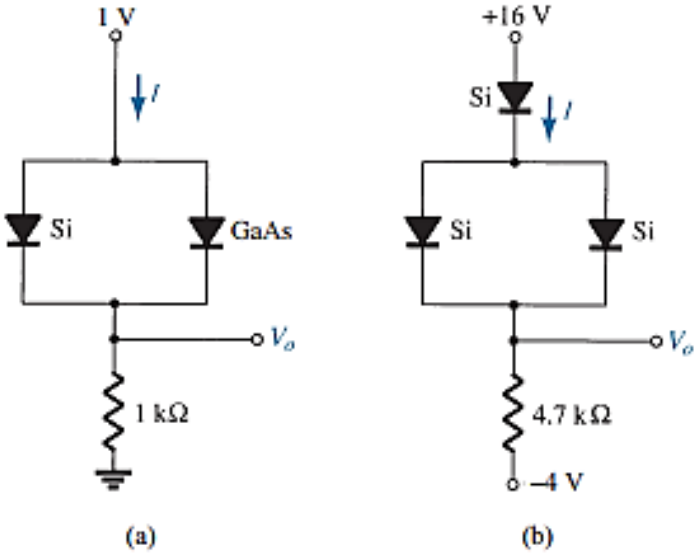
$V_o = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$

(b) Right diode forward-biased:

$$I_D = \frac{20\text{ V} + 4\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} = \mathbf{10.59\text{ mA}}$$

$V_o = 20\text{ V} - 0.7\text{ V} = \mathbf{19.3\text{ V}}$

7 Determine V_o and I for the networks:



(a) Si diode "on" preventing GaAs diode from turning "on":

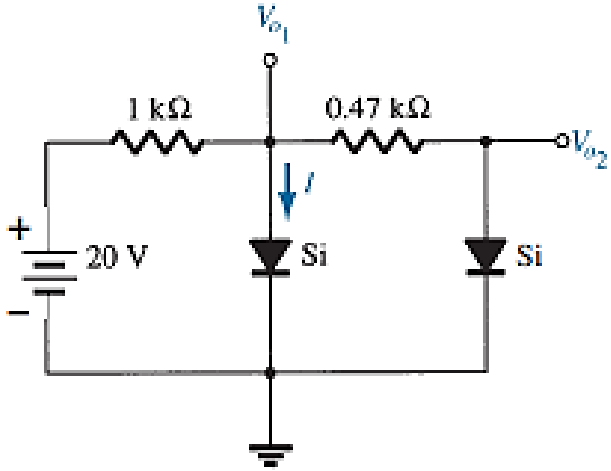
$$I = \frac{1\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \frac{0.3\text{ V}}{1\text{ k}\Omega} = \mathbf{0.3\text{ mA}}$$

$$V_o = 1\text{ V} - 0.7\text{ V} = \mathbf{0.3\text{ V}}$$

(b) $$I = \frac{16\text{ V} - 0.7\text{ V} - 0.7\text{ V} + 4\text{ V}}{4.7\text{ k}\Omega} = \frac{18.6\text{ V}}{4.7\text{ k}\Omega} = \mathbf{3.96\text{ mA}}$$

$$V_o = 16\text{ V} - 0.7\text{ V} - 0.7\text{ V} = \mathbf{14.6\text{ V}}$$

8 Determine V_{o1} and V_{o2} for the network:



Both diodes forward-biased:

$V_{o_1} = \mathbf{0.7\text{ V}}$, $V_{o_2} = \mathbf{0.7\text{ V}}$

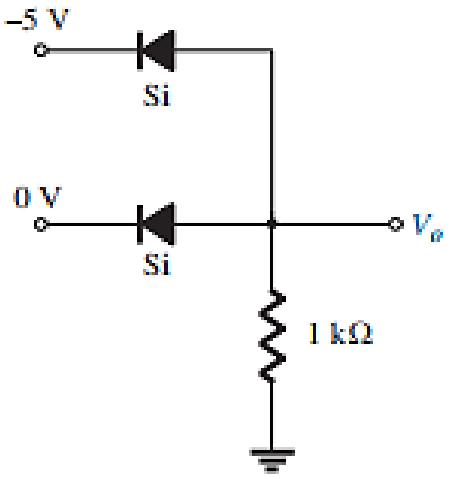
$$I_{1\text{ k}\Omega} = \frac{20\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \frac{19.3\text{ V}}{1\text{ k}\Omega} = 19.3\text{ mA}$$

$I_{0.47\text{ k}\Omega} = 0\text{ mA}$

$I = I_{1\text{ k}\Omega} - I_{0.47\text{ k}\Omega} = 19.3\text{ mA} - 0\text{ mA}$

$= \mathbf{19.3\text{ mA}}$

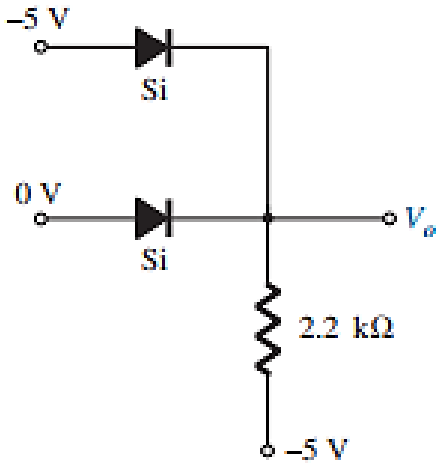
9 Determine V_o for the negative logic OR gate:



The Si diode with −5 V at the cathode is "on" while the other is "off". The result is

$$V_o = -5\text{ V} + 0.7\text{ V} = \mathbf{-4.3\text{ V}}$$

10 Determine V_o for the negative logic AND gate:



0 V at one terminal is "more positive" than −5 V at the other input terminal. Therefore assume lower diode "on" and upper diode "off".
The result:

$$V_o = 0\text{ V} - 0.7\text{ V} = \mathbf{-0.7\text{ V}}$$

The result supports the above assumptions.